THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS FULL SYLLABUS PAPER SET 07

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The 10th triangular number is: 1

(a) 50 **(b)** 45 **(c)** 66 **(d)** 55

2. The sum of the first 6 odd numbers is: 1

(a) 36 **(b)** 30 **(c)** 49 **(d)** 21

3. An angle of 15° is: 1

(a) reflex **(b)** obtuse **(c)** acute **(d)** right

4. An angle is formed by: 1

(a) two line segments of equal length **(b)** a circle

(c) two parallel lines **(d)** two rays with a common starting point

5. The Collatz sequence starting at 7 takes how many steps to reach 1? 1

(a) 16 **(b)** 15 **(c)** 19 **(d)** 17

6. The smallest 4-digit number that can be made from 3, 8, 1, 5 is: 1

(a) 1358 **(b)** 3158 **(c)** 1385 **(d)** 1538

7. The length (height) of a bar shows: 1

(a) the title **(b)** the quantity or frequency

(c) the label **(d)** the colour

8. The key ‘one symbol = 4’ means 6 symbols represent: 1

(a) 20 **(b)** 6 **(c)** 24 **(d)** 10

9. A number is divisible by 6 if it is divisible by: 1

(a) both 2 and 3 **(b)** 2 or 3 **(c)** both 4 and 3 **(d)** 5 and 2

10. Twin primes are primes that differ by: 1

(a) 1 **(b)** 3 **(c)** 4 **(d)** 2

11. The unit of area is: 1

(a) centimetre **(b)** kilogram

(c) square centimetre (cm²) **(d)** litre

12. The area of a rectangle is 72 m² and its length is 9 m. Its breadth is: 1

(a) 7 m **(b)** 8 m **(c)** 9 m **(d)** 63 m

13. $\frac{1}{6} + \frac{1}{4}$ equals: 1

(a) $\frac{1}{2}$ **(b)** $\frac{5}{12}$ **(c)** $\frac{2}{10}$ **(d)** $\frac{11}{30}$

14. $\frac{5}{7} + \frac{1}{7}$ equals: 1

(a) $\frac{5}{7}$ **(b)** $\frac{4}{7}$ **(c)** $\frac{6}{14}$ **(d)** $\frac{6}{7}$

15. A rectangle of sides 6 cm and 4 cm has perimeter: 1

(a) 24 cm **(b)** 20 cm **(c)** 22 cm **(d)** 10 cm

16. In a rectangle, opposite sides are: 1

(a) always 5 cm **(b)** perpendicular
(c) equal and parallel **(d)** unequal

17. Which of these has rotational symmetry of order 2 but not 4? 1

(a) Circle **(b)** Square
(c) Equilateral triangle **(d)** Rectangle

18. How many lines of symmetry does a square have? 1

(a) 1 **(b)** 4 **(c)** 2 **(d)** 8

19. The smallest integer among −8, −2, 0, 3 is: 1

(a) 0 **(b)** −2 **(c)** −8 **(d)** 3

20. Which integer is neither positive nor negative? 1

(a) 10 **(b)** −1 **(c)** 1 **(d)** 0

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 10–15 words each.

21. Is 100 a square number? Give a reason. 2

22. Give an example of mental math to multiply 25 by 8. 2

23. State two rules for drawing a bar graph. 2

24. State the divisibility rule of 11. 2

25. State how a diagonal divides a rectangle. 2

26. What is a unit fraction? Give two examples. 2

27. How can we check parallel lines with a ruler and set square? 2

28. Write the predecessor and successor of −10. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Find the sum $1 + 2 + 3 + 4 + 3 + 2 + 1$ and describe the pattern. 3

30. What are intersecting lines? How many points do they have in common? 3

31. A shop sold 40, 55, 35 and 70 notebooks in four weeks. Choose a scale and write the bar lengths. 3

32. Two triangles of base 8 cm and height 5 cm form a figure. Find the area of the figure if they share the base to make a parallelogram. 3

33. What is the radius of a circle having diameter 12 cm? Use it to draw the circle. 3

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34. Name two digits that look the same in a vertical mirror and one that does not. 3

35. Write all primes less than 30 and state how many there are. 3

OR

Using divisibility tests, check whether 3,456 is divisible by 2, 3, 4, 6 and 9. 3

36. Arrange −9, 6, −2, 0, 4, −5 in descending order. 3

OR

Add $(-7) + (-5)$ and $(-7) + 5$ and explain the difference. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. (i) What are collinear points? 5
(ii) How many lines can pass through 2 points?
(iii) How many through 3 non-collinear points taken in pairs?

OR

(i) Find the angle through which the hour hand of a clock turns from 3 p.m. to 7 p.m. 5
(ii) the minute hand from 3:00 to 3:40.

38. **Case study:** A school bell rings every 15 minutes and a club bell every 20 minutes starting at 9:00. 5
(i) After how many minutes do they ring together?
(ii) At what time?
(iii) How many times between 9:00 and 12:00?
(iv) Which number idea is used?

OR

(i) Explain mental maths strategies for addition. 5
(ii) Compute 99 + 76 mentally.
(iii) Compute 5 × 48.

39. (i) Divide 7 equal cakes among 3 friends. 5
(ii) Write each share as a mixed fraction.
(iii) Explain with a picture.

OR

(i) Mention one historical fact about fractions in India. 5
(ii) Write 3/4 in Indian number words.
(iii) Why were fractions needed?

40. (i) Define line symmetry. 5
(ii) Find the lines of symmetry of a square, an equilateral triangle and a regular pentagon.
(iii) Draw them.

OR

Case study: A rangoli design is a regular hexagon with six petals. 5
(i) How many lines of symmetry?
(ii) Order of rotation?
(iii) Smallest angle?
(iv) Is it symmetrical about the centre?

END OF QUESTION PAPER — Total: 80 Marks

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